

HDMoE: A Hierarchical Decoupling-Fusion Mixture-of-Experts Framework for Multimodal Cancer Survival Prediction

Paper ID 3464

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Abstract. Multimodal survival prediction, a crucial yet challenging task, demands the integration of multimodal medical data (*e.g.*, Whole Slide Images (WSIs) and Gene Profiles) to enable accurate survival prediction. Due to the heterogeneity between modalities, methods based on feature decoupling-fusion paradigm have become mainstream. However, these methods have the following shortcomings: (1) fail to reduce the redundant information of modality features before decoupling, which negatively affects the feature decoupling and fusion effect; (2) lack the ability to model the fine-grained relationships of the features and capture the local information interactions between intra- and inter-modality features. To address these issues, we propose a Hierarchical Decoupling-Fusion Mixture-of-Experts (HDMoE) framework with two levels of MoE and Random Feature Reorganization (RFR) modules. In the first-level MoE, shared experts and routed experts are employed to remove redundant information and extract fine-grained specific features within each modality, while the second-level MoE facilitates fine-grained inter-modality feature decoupling. Besides, we design two RFR modules following each level of MoE to finely fuse intra- and inter-modality features, which can help the model capture more fine-grained relationships between modalities. Extensive experimental results on our private Liver Cancer (LC) and three TCGA public datasets confirm the effectiveness of our proposed method. The code will be made publicly available.

Keywords: Multimodal Learning · Mixture-of-Experts · Survival Prediction

1 Introduction

Cancer survival analysis plays a crucial role in understanding patient prognosis and guiding clinical decision-making. Recent studies have shifted focus from unimodal prediction approaches [7,15,18,9] to leveraging more complex multimodal data, thereby enhancing predictive accuracy and capabilities [2,29,23,22]. Taking WSIs and Gene Profiles as an example, WSIs provide spatial organization and morphological features of tumor tissues, while Gene Profiles reveal molecular-level variations, such as mutations and gene expression changes, making them

complementary in nature [14]. However, the effective fusion of these data poses substantial challenges, particularly due to modality heterogeneity and the complex learning of modality relationships [11,1,16].

Known studies have attempted to alleviate the above problems using cross-attention [29,2,23], MOE-based progressive fusion [22], and modality decoupling-fusion approaches [13,30,27]. However, these methods have not fully considered (1) *redundancy within the modality data* and (2) *fine-grained relationships between the modality features*. Here, redundant information refers to repetitive or highly correlated features within modality representations [24,8]. For instance, WSI patches often exhibit similarities in tissue structures across different regions, while Gene Profiles within the same pathway may display high correlations. Consequently, effectively reducing redundant information can enhance the model’s capacity to capture crucial insights within complex modality relationships. Concerning the second point, although there exists a fine-grained correspondence between specific gene expressions and local regions of WSIs [28] (*e.g.*, the relationship between specific gene mutations and the tumor microenvironment), most works [13,30,27] still learn relationships at the level of global features, making it difficult to fully capture local interactions within and across modalities.

To address these challenges, we propose HDMoE, a Hierarchical Decoupling-Fusion Mixture-of-Experts framework, which employs an interleaved design consisting of two levels of identical Sparse MoE [6,17] and RFR modules, following a purify-then-decouple strategy. Specifically, modality features are first divided into multiple tokens as input to enable fine-grained feature relationship learning. In the first-level MoE, we introduce a shared expert [3] to extract unimodal fine-grained generic features, eliminating the redundant intra-modality features. Additionally, routed experts are employed to extract unimodal specific features at the token level. To help the model capture more local feature relationships, purified and specific features from different modalities are randomly reorganized at a fine-grained level. The second-level MoE is then applied for inter-modality feature decoupling. The reorganized fusion features are first divided into multiple fine-grained tokens, after which fine-grained inter-modality shared and specific decoupled features are extracted by the shared and routed experts, respectively. Finally, the decoupled features are randomly reorganized, which can diversify the fine-grained relationships between decoupled features during fusion. Overall, the MoE with shared expert serves to reduce redundancy in modality features, while the Sparse MoE and RFR provide the model with the ability to explore fine-grained feature relationships both within and across modalities. Our main contributions are summarized as follows:

- We target the problem of modality-decoupling-fusion-based survival analysis and propose HDMoE to refine modality features before decoupling and effectively capture fine-grained intra- and inter-modality relationships.
- We design two RFR modules following each MoE level. These modules achieve feature fusion while assisting the model in capturing more fine-grained feature relationships by feature segmentation and reorganization.

- We conduct extensive experiments and ablation studies on multiple datasets for cancer survival analysis, demonstrating that our proposed method achieves state-of-the-art performance.

2 Methodology

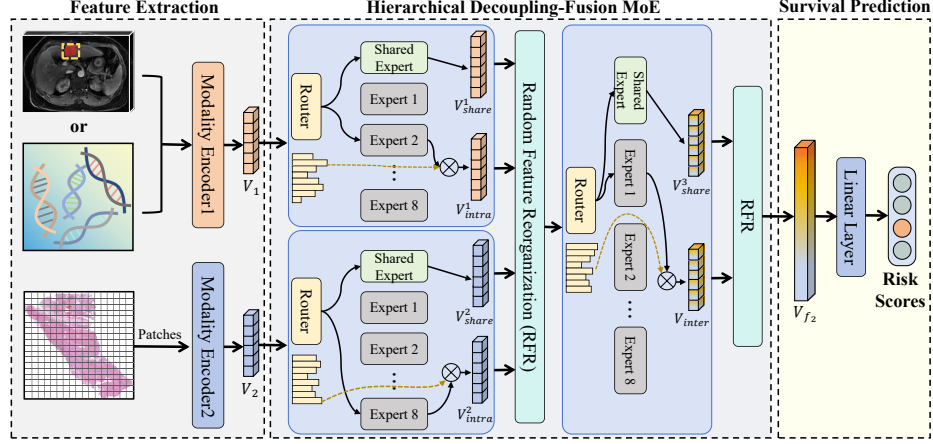


Fig. 1. An overview of our proposed framework, consisting of three modules: Feature Extraction, Hierarchical Decoupling-Fusion MoE, and Survival Prediction.

2.1 Unimodal Feature Extraction

As illustrated in Fig. 1, our pipeline initiates by extracting unimodal representations from the patient’s WSIs alongside either MRI scans or Gene Profiles. Specifically, due to the massive resolution of WSIs, following the common setting [29,2,23,30], we adopt the CLAM [15] to divide tissue regions into non-overlapping 256×256 patches at a $20\times$ magnification level. We then utilize an ImageNet pre-trained ResNet50 [5] to extract patch features, represented as $P \in \mathbb{R}^{n_p \times d}$, where n_p is the number of patches in WSIs and d is the dimensionality of the modality latent features. For MRI scans, we first crop the tumor-relevant regions based on tumor segment labels, and then also employ ResNet50 to extract spatial and sequential features of the tumor. After that, global average pooling is applied to obtain general MRI features, resulting in $M \in \mathbb{R}^{n_m \times d}$, where n_m is the number of MRI features. For genomic data, we extract genomic sub-sequence features following SNN [9], represented as $G \in \mathbb{R}^{n_g \times d}$, where n_g is the number of genomic sub-sequences. After that, the learned modality-aware class tokens ($V_1, V_2 \in \mathbb{R}^{1 \times d}$ represents P and M or G , respectively) derived from Attention [18] are viewed as modality representations, which aim to equalize the number of different modality features.

2.2 Dual-Level Mixture of Decoupled Experts

To enhance the model’s capability for fine-grained feature learning, we configure the Sparse MoE as the core component of our framework. This design not only enables multi-expert and multi-perspective feature learning but also flexibly captures local detailed information. Building on this, we propose a hierarchical MoE framework that follows a “purify first, then decouple” strategy:

First-Level MoE. We configure shared and routed experts to refine intra-modality features from global and local perspectives, respectively. The shared expert captures the common information across all token features by sharing the parameters between feature tokens (*e.g.*, gene pathway activity or tissue architecture features in WSIs), reducing redundant information. Besides, routed experts extract modality-specific features (*e.g.*, key mutations associated with particular cancer subtypes or localized anomalous regions in WSIs). Specifically, we first divide feature V_i into T tokens $\{v_t^i\}_{t=1}^T$ ($i \in \{1, 2\}$). Given $N + 1$ experts $\{E_1^i, \dots, E_N^i, E_{share}^i\}$ and a router G_{intra}^i that activates the Top-K experts:

$$\begin{aligned} G_{intra}^i(v_t^i) &= W_{intra}^i \cdot v_t^i, \quad g_{intra}^i(v_t^i) = \text{Top-K}(\text{Softmax}(G_{intra}^i(v_t^i))), \\ V_{intra}^i &= \text{Concat}\left(\{g_{intra}^i(v_t^i) \cdot E_j^i(v_t^i)\}_{t=1}^T\right), \quad V_{share}^i = E_{share}^i(\{v_t^i\}_{t=1}^T), \end{aligned} \quad (1)$$

where W_{intra}^i is a learnable weight matrix for feature v_t^i , g_{intra}^i denotes the routing score, and j denotes the selected expert index from g_{intra}^i .

Second-Level MoE. The second-level MoE is designed to achieve inter-modality feature decoupling and adopts the same architecture as the First-level MoE. The shared and routed experts are utilized to capture fine-grained shared features (*e.g.*, the correlations between gene mutation patterns and histopathological morphology) and specific features (key biomarkers within each modality), for decoupling purposes. Specifically, we use v_{f_1} represents one of T tokens in V_{f_1} (obtained by RFR, see Eq. (3) below). Given another $N + 1$ experts $\{E_1^{inter}, \dots, E_N^{inter}, E_{share}^3\}$ and a router G_{inter} :

$$\begin{aligned} G_{inter}(v_{f_1}) &= W_{inter} \cdot v_{f_1}, \quad g_{inter}(v_{f_1}) = \text{Top-K}(\text{Softmax}(G_{inter}(V_{f_1}))), \\ V_{inter} &= \text{Concat}\left(\{g_{inter}(v_{f_1}) \cdot E_j^{inter}(v_{f_1})\}_1^T\right), \quad V_{share}^3 = E_{share}^3(\{v_{f_1}\}_1^T), \end{aligned} \quad (2)$$

where W_{inter} is a learnable weight matrix, g_{inter} denotes the routing score, and j denotes the selected expert index from g_{inter} .

2.3 Random Feature Reorganization (RFR)

To facilitate the model in capturing more information about the relationships among fine-grained features, we design a RFR module. This module is capable of achieving feature fusion while further diversifying the combinations of fine-grained features. The process can be formulated as Eq. (3),

$$V_{f_1} = \text{RFR}_1(\{V_{intra}^1, V_{share}^1, V_{intra}^2, V_{share}^2\}), \quad V_{f_2} = \text{RFR}_2(\{V_{inter}, V_{share}^3\}), \quad (3)$$

where RFR_1 is the first-level RFR following the first-level MoE, positioned to enhance the second-level MoE’s capacity for capturing fine-grained local features, while the RFR_2 is deployed downstream of the second-level MoE, specifically designed to optimize feature diversification by establishing heterogeneous relational patterns among the refined feature representations. Specifically, we first set n segment values $S = \{s_k\}_{k=1}^n$ to divide each MoE output V_o into L equal feature segments $\{v_{o,l}\}_{l=1}^L$, where $L = d/s_k$, V_o represents any one of $\{V_{intra}^1, V_{share}^1, V_{intra}^2, V_{share}^2, V_{inter}, V_{share}^3\}$. After that, each feature segments is reorganized. Taking V_{f_2} as an example, the process can be formulated as Eq. (4),

$$V_{f_2} = \text{Concat}(\{V_l'\}_{l=1}^L), V_l' = \text{Concat}(\{v_{inter,l}, v_{share,l}^3\}). \quad (4)$$

Notably, the segment value is randomly chosen at every forward step.

2.4 Survival Prediction

In survival prediction module, we use one fully connected layer to estimate the hazard function $h(t) = h(T = t \mid T \geq t, (V_1, V_2)) \in [0, 1]$, which is the probability of death for a patient right after the time point t .

To achieve decoupling, we employ a Distance Metric (DM) function to control the relationships between different features. Specifically, output features within each MoE level are kept apart, while features across MoE level are kept closer, as shown in Eq. (5). Functions DM and DM' are inversely related to each other.

$$\begin{aligned} \mathcal{L}_{\text{dm}} = & \sum_{i=1}^2 \text{DM}'(V_{intra}^i, V_{share}^i) + \text{DM}'(V_{inter}, V_{share}^3) \\ & + \text{DM}(\{V_{intra}^i\}_{i=1}^2, V_{inter}) + \text{DM}(\{V_{share}^i\}_{i=1}^2, V_{share}^3). \end{aligned} \quad (5)$$

Additionally, to prevent the majority of tokens from being routed to a small subset of experts, which could lead to insufficient training of the remaining experts [19], we introduce a load balance loss [10,21] as follows:

$$\mathcal{L}_{\text{bl}} = \sum_{i=1}^N f_i P_i, f_i = \frac{1}{T} \sum_{t=1}^T \mathbb{1}(v_t \text{ selects Expert } j), P_i = \frac{1}{T} \sum_{t=1}^T \text{Softmax}(W \cdot v_t), \quad (6)$$

where T denotes the number of tokens, $\mathbb{1}(\cdot)$ is the indicator function, and W represents learnable weight from gating network. The total loss is shown in Eq. (7),

$$\mathcal{L} = \mathcal{L}_{\text{surv}} + \alpha \mathcal{L}_{\text{dm}} + \beta \mathcal{L}_{\text{bl}}, \quad (7)$$

where $\mathcal{L}_{\text{surv}}$ is the survival prediction loss [25], α and β are balance factors.

3 Experiments and Results

3.1 Datasets

We conduct extensive experiments on our LC dataset as well as three public TCGA¹ (The Cancer Genome Atlas) datasets to showcase the robust perfor-

¹ <https://portal.gdc.cancer.gov>

Table 1. C-Index (mean $\pm$ std) performance over our LC dataset. Img. and Patho. refer to Imaging modality (MRIs) and pathological modality (WSIs). The best results are underlined in red, while the second best are italicized in *blue*, respectively.

Model	Modality Img. Patho.	C-index	Model	Modality Img. Patho.	C-index
ResNet3D [5]	✓	0.645 $\pm$ 0.036	MCAT [2]	✓ ✓	0.614 $\pm$ 0.040
MaxMIL [20]	✓	0.570 $\pm$ 0.051	CFDL [13]	✓ ✓	0.612 $\pm$ 0.053
MeanMIL [26]	✓	0.500 $\pm$ 0.051	CCL [30]	✓ ✓	0.648 $\pm$ 0.034
AttnMIL [7]	✓	0.520 $\pm$ 0.052	CMTA [29]	✓ ✓	0.597 $\pm$ 0.079
CLAM_MB [15]	✓	0.510 $\pm$ 0.053	MOTCat [23]	✓ ✓	0.621 $\pm$ 0.043
CLAM_SB [15]	✓	0.518 $\pm$ 0.041	MoME [22]	✓ ✓	<i>0.650</i> $\pm$ 0.037
TransMIL [18]	✓	0.592 $\pm$ 0.047	HDMoE (ours)	✓ ✓	<u>0.683</u> $\pm$ 0.029

mance of our HDMoE. The LC Dataset comprises 160 pairs of diagnostic MRIs and WSIs annotated with ground-truth survival outcomes. Each MRIs cover five distinct sequences per patient: A, D, PMX, T1, and T2, while the WSIs are stained using H&E method [12]. The TCGA datasets include paired WSIs and genomic data. We use three typical cancer types: Bladder Urothelial Carcinoma (BLCA, n=373), Breast Invasive Carcinoma (BRCA, n=956), and Lung Adenocarcinoma (LUAD, n=453). Consistent with prior studies [2], the genomic data are categorized into six sequences: Tumor Suppression, Oncogenesis, Protein Kinases, Cellular Differentiation, Transcription, and Cytokines and Growth.

3.2 Implementation Details

Training Settings. We compare HDMoE with unimodal and multimodal baselines, including: SNN [9], SNNTrans [9], MaxMIL [20], MeanMIL [26], AttnMIL [7], CLAM [15], TransMIL [18], MCAT [2], CMTA [29], MOTCAT [23], CFDL [13], PIBD [27], CCL [30] and MoME [22]. For LC dataset, ResNet3D [5] is used as MRIs-related unimodal baseline model. The C-Index [4] is adopted as the evaluation metric. In our experiments, we perform 5-fold cross-validation.

Hyper-parameters. For the proposed HDMoE, we implement the framework by setting $d = 256$, $n_m = 1$, and $n_g = 6$. We use SGD optimizer with a learning rate of 1×10^{-3} for LC dataset and Adam optimizer with a learning rate of 5×10^{-4} for TCGA datasets. The batchsize is set to 1, weight decay is set to 1×10^{-3} , and the number of epochs is 30. After optimizing the hyper-parameters, we set the balance factors $\alpha = 1$, $\beta = 0.01$, Tokens $T = 8$, numbers of experts $N = 8$, Top-K = 1 and segment values $S = \{1, 2, 4, 8, 16, 32, 64, 128\}$.

3.3 Comparison Results

On LC Dataset, results show that our method outperforms all other methods, see Tab. 1. For example, in CCL method ($\downarrow$ 3.5%), redundant information in the features is not removed before feature decoupling, and the MoME method

Table 2. C-Index (mean $\pm$ std) performance over three TCGA datasets. Genomic and Pathology refer to Genomic modality (Genomic profiles) and Pathological modality (WSIs), respectively. The best results are underlined in **red**, while the second best are italicized in **blue**, respectively. * represents we rerun the method.

Model	Modality		Dataset			
	Geno.	Patho.	BLCA	BRCA	LUAD	Overall
SNN [9]	✓		0.618 $\pm$ 0.022	0.633 $\pm$ 0.074	0.611 $\pm$ 0.047	0.621
SNNTrans [9]	✓		0.659 $\pm$ 0.032	0.648 $\pm$ 0.058	0.638 $\pm$ 0.022	0.648
MaxMIL [20]		✓	0.551 $\pm$ 0.032	0.600 $\pm$ 0.055	0.596 $\pm$ 0.060	0.582
MeanMIL [26]		✓	0.585 $\pm$ 0.032	0.611 $\pm$ 0.029	0.576 $\pm$ 0.054	0.591
AttnMIL [7]		✓	0.599 $\pm$ 0.048	0.590 $\pm$ 0.047	0.620 $\pm$ 0.061	0.603
CLAM_MB [15]		✓	0.565 $\pm$ 0.027	0.620 $\pm$ 0.052	0.582 $\pm$ 0.072	0.589
CLAM_SB [15]		✓	0.559 $\pm$ 0.034	0.609 $\pm$ 0.033	0.594 $\pm$ 0.063	0.587
TransMIL [18]		✓	0.575 $\pm$ 0.034	0.643 $\pm$ 0.037	0.642 $\pm$ 0.046	0.620
MCAT [2]	✓	✓	0.672 $\pm$ 0.032	0.659 $\pm$ 0.042	0.659 $\pm$ 0.027	0.663
CMTA* [29]	✓	✓	0.660 $\pm$ 0.026	0.652 $\pm$ 0.058	0.669 $\pm$ 0.033	0.660
MOTCat* [23]	✓	✓	0.673 $\pm$ 0.037	0.673 $\pm$ 0.032	0.661 $\pm$ 0.039	0.669
CFDL* [13]	✓	✓	0.675 $\pm$ 0.025	0.658 $\pm$ 0.038	0.664 $\pm$ 0.030	0.666
PIBD* [27]	✓	✓	0.599 $\pm$ 0.039	0.643 $\pm$ 0.029	–	0.621
CCL* [30]	✓	✓	0.640 $\pm$ 0.041	0.669 $\pm$ 0.043	0.657 $\pm$ 0.037	0.655
MoME* [22]	✓	✓	0.674 $\pm$ 0.020	0.679 $\pm$ 0.035	0.672 $\pm$ 0.056	0.675
HDMoE (Ours)	✓	✓	0.694 $\pm$ 0.019	0.686 $\pm$ 0.022	0.675 $\pm$ 0.014	0.685

($\downarrow$ 3.3%) fails to fully consider the relationship between the fine-grained modality features, and instead uses the overall features as inputs for relationship learning. These results demonstrate the effectiveness of our method, particularly the first-level MoE, which effectively reduce redundant information from MRIs and WSIs. Additionally, the Sparse MoE and RFR modules enhance the model’s capability to learn fine-grained relationships between modality features.

Results on three TCGA Datasets are shown in Tab. 2. Our method demonstrates outstanding performance, especially on BLCA and BRCA datasets. Specifically, compared to MoME method, our method achieves improvements of 2.0%, 0.7%, and 0.3% on three datasets and a 1% improvement in overall performance. These results indicate that our method is also well-suited for integrating genomic and pathological data. Notably, genomic-based methods generally outperform pathology-based methods and even some multimodal approaches, highlighting the critical role of genomic modality and the importance of effectively integrating genomic and pathological data. By hierarchical MoE and RFR modules to eliminate modality redundancies and enhance fine-grained inter-modality interactions, our method improves the efficacy of feature decoupling-fusion paradigm.

Table 3. Ablation study on LC and BLCA datasets to evaluate the effectiveness of Feature Reorganization (FR.), Balance loss ($\mathcal{L}_{bl}$), Decouple loss ($\mathcal{L}_{dm}$), Fine-grained feature (Ff.), De-redundancy (Dr.) and the different choices of Distance Metric (DM).

Model	Settings					Dataset		DM	Dataset	
	FR.	$\mathcal{L}_{bl}$	$\mathcal{L}_{dm}$	Ff.	Dr.	LC	BLCA		LC	BLCA
HDMoE	✓			✓	✓	0.654 ± 0.012	0.665 ± 0.026	L1	0.647 ± 0.027	0.659 ± 0.048
HDMoE	✓	✓		✓	✓	0.657 ± 0.016	0.670 ± 0.036	KL	0.652 ± 0.031	0.648 ± 0.013
HDMoE		✓	✓	✓	✓	0.640 ± 0.036	0.670 ± 0.022	MSE	0.680 ± 0.042	0.644 ± 0.048
HDMoE	✓		✓	✓	✓	0.659 ± 0.019	0.681 ± 0.011	COS	0.683 ± 0.029	0.694 ± 0.019
HDMoE	✓	✓	✓		✓	0.653 ± 0.025	0.666 ± 0.025	/	/	/
HDMoE	✓	✓	✓	✓		0.655 ± 0.026	0.670 ± 0.028	/	/	/
HDMoE	✓	✓	✓	✓	✓	0.683 ± 0.029	0.694 ± 0.019	/	/	/

3.4 Ablation Studies

Settings of Framework Components. We conduct ablation experiments to evaluate the effectiveness of the framework components. Note that we accomplish the fine-grained and de-redundancy ablation experiments by not dividing the features V_1, V_2 into multiple tokens and by substituting V_1, V_2 for the purified features output by the first-level shared experts. Results are shown in the left part of Tab. 3. We observe that removing any one or two components leads to performance degradation. For instance, the lack of fine-grained and de-redundancy components will bring 2.8%, 2.4% performance degradation on BLCA dataset, which also illustrates the importance of exploring inter-modality fine-grained feature relationships and removing feature redundancies before decoupling.

Impact of Distance Metrics. We investigate the effect of different distance metrics used in decoupling loss. By default, we employ COS (Cosine Similarity) to measure the distance between decoupled feature representations. Furthermore, we compared COS with alternative metrics, including L1 norm, KL (Kullback-Leibler) divergence, and MSE (Mean Squared Error). As shown in the right part of Tab. 3, using COS distance metric empirically produces better results.

4 Conclusion

In this paper, we proposed a HDMoE framework for multimodal cancer survival analysis. The first-level MoE is able to reduce intra-modality redundancies and extract token-level specific features within each modality. The second-level MoE further captures complementary information between modalities and obtains inter-modality shared and specific features. Besides, RFR module was introduced to enrich inter-modality relationships while achieving feature fusion. Extensive experiments demonstrated that our method outperforms other methods, making it applicable to survival analysis tasks involving different types of modalities.

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